

Mining Attention Dynamics: Auditing KV-Saliency Prediction in LLMs

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Abstract—Deciding which cached attention keys and values (KV) to retain requires evaluating predictions under the budget and candidate set of each retention decision. KVSaliencyBench audits these evaluation objects through future-attention labels, explicit selector semantics, source-prompt splits and versioned code, and scores pooled and per-decision metrics together. On corrected $\leq 4K$ -token traces from two instruction-tuned models, a three-parameter logistic scorer reaches pooled area under the receiver operating characteristic curve (AUC) 0.890 against LightGBM’s 0.896, a gap of at most 0.008 across tree capacities and training sizes. Yet the within-layer attention signal recovers 0.659 of each decision’s future top-decile blocks, versus 0.613 and 0.615 for those models. Pooled AUC is almost entirely cross-decision pairs: the logistic scorer’s binary cross-layer term raises it by 0.017 but lowers within-decision AUC and recall, and the reversal narrows at 20% retention. A pinned bfloat16 masked-loop rerun finds a runtime-reconstructed scorer and an H2O-style accumulator equivalent in mean LongBench F1 within a practical margin of two absolute points over seven datasets, both below full cache. A feature bridge leaves a historical served-oracle gap unexplained, and its on-policy arm ranks only surviving blocks. A physical reference backend stores only the retained KV, but extended mask-versus-compaction replay exceeds its numerical tolerance, so no serving equivalence is claimed. These bounded findings support evaluating prediction at the selector’s operating point, not a serving or per-task equivalence claim.

Index Terms—KV cache, attention prediction, empirical audit, calibration, benchmarking, closed-loop evaluation.

I. INTRODUCTION

Cached attention keys and values (KV state) grow with context while attention concentrates on few positions. KV-management methods exploit this through accumulated attention [1], attention sinks with recent windows [2], recent-query observations [3], query-key page bounds [4], cross-layer information [5] and learned retention [6]. Most rest on a prediction—that the blocks kept are the ones future queries will attend to—and learned predictors now target attention directly [7]. What is rarely separated is pooled ranking, the per-decision budget a selector imposes, the features a runtime can reconstruct, the candidate set and task quality. Methods are

mostly compared at the end of the pipeline, on answer scores, where the scorer, the budget rule, the features and the feedback trajectory all move together, so a gain cannot be attributed to any one. Better prediction need not identify a better retention policy: selection changes later model states and their outputs.

Measuring the prediction itself needs objects a task benchmark does not record: the candidate set and budget of each retention decision, the features available at runtime, and the future attention the decision is betting on. KVSaliencyBench records them, so that a gain can be attributed to prediction rather than to a budget rule and a proxy ranking checked against task quality.

Contributions. (1) *A protocol and benchmark.* KVSaliencyBench fixes future-attention labels, selector semantics (Table I), request- and source-prompt splits, tie-aware per-decision metrics and versioned collection; its default entry point (version 2.0) scores pooled and per-decision metrics together from a prediction table alone. (2) *Controlled findings.* A three-parameter logistic scorer on two attention-magnitude views is within 0.006 AUC of LightGBM on v2 traces (0.008 at most across tree capacities) without a calibration step, yet at the per-decision top-decile budget the within-layer signal alone recovers more future top-decile blocks than either, a reversal that shrinks at 20% retention; an exact split shows pooled AUC is a cross-decision statistic, which the scorer’s binary cross-layer term raises while lowering within-decision AUC and recall. A pinned bfloat16 masked-loop rerun finds the reconstructed scorer and the H2O-style accumulator mean-F1 equivalent within a practical ± 0.02 margin over seven datasets (two one-sided tests, TOST, $p = 0.0023$), and a bridge measures feature reconstruction (§IV-B). (3) *An audit and its repair.* We diagnose corrupted float16 Qwen2.5 generation, rerun with a pinned simulator, audit capacity effects from logged counts, and report physical-KV parity levels.

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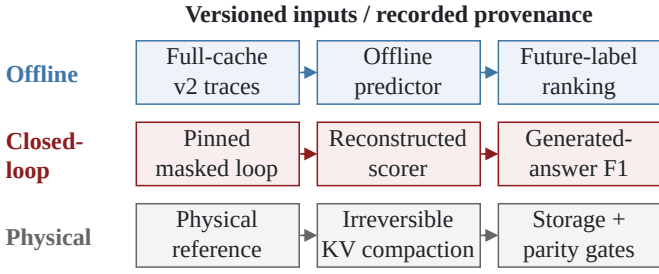


Fig. 1. The three evaluation paths, one per row: offline prediction (blue), closed-loop task quality (red) and the physical reference (grey); their selector semantics differ (Table I).

II. PROBLEM AND EVIDENCE LEVELS

A. Attention-defined saliency

At layer ℓ and decode step t , let $\mathcal{B}_{\ell,t}$ contain the logical KV blocks currently under consideration. A block groups contiguous original token positions. We aggregate attention by averaging over heads and over a block’s token positions; denote the resulting statistic by $a_{\ell,b,t}$. It is a block-mean attention statistic, not necessarily a mass normalized to sum to one over blocks. For a horizon h and a fraction r , the target is $z_{\ell,b,t}^{(h)} = \mathbf{1}[b \in \text{Top}_{\lceil r|\mathcal{B}_{\ell,t}| \rceil} \{a_{\ell,b',t+h} : b' \in \mathcal{B}_{\ell,t}\}]$: ranking is restricted to blocks present at t . We use $r = 0.1$ and report $h = 4$; horizons 1, 16 and 64 are recorded only.

A scorer $f(\mathbf{x}_{\ell,b,t})$ estimates the label probability from recorded features; examples arrive in decode order, positives are a small fraction of blocks, drift is a hypothesis to test, and serial dependence means millions of block rows are not millions of independent observations.

B. Three different evaluation targets

Fig. 1 locates the evaluation paths. Offline prediction scores future-attention labels on fixed full-cache traces; served-oracle diagnostics score a current-attention target along the evaluated trajectory, and attention importance is not answer correctness; closed-loop task quality scores answers the selector’s own decisions helped produce, here through masking rather than a cache manager. A fraction of blocks allowed to attend is a *logical retention budget*, not of device memory.

C. Statistical protocol

Offline ranking uses AUC, threshold-based AUPRC, precision/recall at $k = r$, Brier score and, for scorers that emit probabilities only, expected calibration error (ECE). Pooled metrics rank rows *across* decisions; top- k metrics are reported pooled over test rows *and* per cache decision (request, layer, step) at the label’s own $\lceil rn \rceil$, and pooled AUC is split exactly into same-decision and cross-decision pairs. Training and testing use disjoint requests and paired bootstrap intervals resample requests, not block rows. Unless marked otherwise, intervals are 95%; equivalence intervals are 90%. A 10,000-draw sensitivity analysis jointly resamples held-out model-requests that share a source prompt, on both splits. Top- k score ties use a stable descending sort, and v2 label boundaries

keep the collector’s `argpartition` rule with exactly $\lceil rn \rceil$ positives; neither is randomized, the artifact states both orders exactly, and recomputing every per-decision recall under exact random tie-breaking moves none by more than 2×10^{-5} . The main downstream comparison aggregates paired request scores into architecture–dataset cells and averages the two architectures within each dataset; a TOST analysis [8], [9] asks whether the mean difference over the seven datasets lies inside a practical margin. Failing to detect a difference is not equivalence, and two comparisons equivalent to one baseline need not be equivalent to each other.

III. SIGNALS AND EMPIRICAL FEATURE SELECTION

A. Recorded views

The feature family contains a normalized within-layer exponential moving average (EMA) of attention $x_{wl} = \bar{a}_{\ell,b,t} / \max_{b'} \bar{a}_{\ell,b',t}$, a previous-layer hot-block indicator $x_{cl} = \mathbf{1}[\bar{a}_{\ell-1,b,t} \text{ is in the top-}r]$, a query-derived proxy $x_q = (1 + \cos(\tilde{q}_t, \tilde{K}_{\ell,b})) / 2$ and a temporal-age feature $x_{age} = \exp(-(t - p_b)/w)$. Here $\bar{a}$ is an EMA and p_b the block-creation step, so x_{age} is an age proxy, not an LRU access-recency measurement; the tildes mark the archived query representation, whose coordinate-space audit in §IX limits conclusions about that view. The cross-layer signal is an attention-magnitude carry-over, not InfiniGen’s query-based mechanism, and the offline extractor falls back to the current layer’s own EMA at layer 0 where runtime realizations need explicit fallback and alignment rules (Table I).

B. Relevance diagnostics

We estimate relevance $I(X_i; Z)$ with quantile-binned plug-in estimators, in nats. On the version-2 corpus, within-layer EMA and the cross-layer indicator have $I(X_i; Z)$ of 0.104 and 0.099 nats, versus 0.007 for the cosine query proxy and under 0.001 for age (§V-A); raw-probe AUCs are in Table II.

Archived conditional-information diagnostics $U_i = I(X_i; Z \mid X_{-i})$ agree that the magnitude views carry what the proxies do not (≤ 0.006 nats): within-layer is 0.133–0.155 on every binning grid and the binary cross-layer estimate 0.048–0.084 on the finer ones, collapsing to zero under the coarsest, so the artifact reports its grid sensitivity. These estimates motivate a candidate subset, evaluated on held-out requests; they are not a sufficiency certificate, and the evidence for the compact model is its measured predictive performance.

IV. PREDICTOR FAMILY AND FEATURE REALIZATIONS

A. A compact offline scorer

We fit a ridge-regularized logistic model by Newton–IRLS, $f(\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b)$; on the within-layer and cross-layer features it has two coefficients and one intercept. A log-loss fit without class reweighting has low measured ECE here and the class-balanced LightGBM does not, but an unweighted or recalibrated tree removes that difference (§V): calibration is separate from ranking here.

B. Offline predictor versus runtime reconstruction

The masking-loop scorer reconstructs the cross-layer view from the previous layer’s EMA, neutralizes the query view, and feeds selection into later states, so its task F1 cannot inherit the offline predictor’s AUC. Table I distinguishes fixed offline features, policy-dependent runtime features and the irreversible physical reference. **Bridge experiment.** With the frozen archived two-view checkpoint, one label (top- $[0.1n]$ of block attention four steps ahead among the blocks present now and still present then) and the 128 frozen prompts of §VI, we score (A) offline-style features re-derived from the full-cache attention stream under version-2 conventions, (B) the reference backend’s runtime realization on the same rows, and (C) that realization along the on-policy trajectory at 20% budget. In A and B the candidates are every block present; in C they are the blocks retained now *and* at $t + h$, a survivor-conditioned arm whose n and label denominator are taken among those survivors. Mean per-request AUC is 0.902/0.899/0.989 on Llama-3.1-8B (bfloat16) and 0.894/0.892/0.989 on Qwen2.5-7B (float32): the feature realization costs $-0.003 [-0.004, -0.001]$ (84/128 requests negative) and $-0.002 [-0.003, -0.000]$ (75/128), whereas the on-policy arm is higher by $+0.090$ and $+0.098$ (128/128 both). That is not evidence that retention makes prediction easier: C ranks only blocks the policy already kept, so an evicted block can never be missed, and the value neither isolates a feedback effect nor measures the coverage lost to eviction. Neither contrast reproduces the archived 0.948-to-0.643 drop of §VIII, which also differs in corpus, aggregation and numerics; attribution would need a label-only arm on identical trajectories, candidates, features and scorer, so that gap stays unexplained.

C. Selector semantics across the three evidence objects

Table I lists the rules that decide what each object measures. Two differ on purpose: the masked loop (pinned-rerun rules) recomputes its kept set every eight steps, so an excluded block can return, and budgets $\text{round}(0.2n)$ blocks with n counting the blocks of generated tokens too, whereas the reference deletes irreversibly and fixes its budget from the prompt.

D. Thresholds and online updates

A score can drive a fixed-size top- k set or a thresholded set $S_{\ell,t} = \{b : f(\mathbf{x}_{\ell,b,t}) \geq \tau_\ell\}$. The former fixes logical retention, whereas the latter lets set size vary. We evaluate static and online variants separately. Threshold results are reported in §VIII; its empirical in-distribution performance is not a hard per-request risk guarantee.

V. PREDICTION MEASUREMENTS

Data and comparators. Corrected version-2 traces carry the offline evidence; version-1 diagnostics are in §VIII. Single-signal rows score one recorded feature and do not reimplement the papers that motivate them; the learned comparators are LightGBM on all four views (150 trees of depth 3) and the two-view logistic fit, with larger logistic, MLP and online fits

TABLE I
SELECTOR AND LABEL SEMANTICS. BLOCKS ARE 32 TOKENS; n IS THE CANDIDATE COUNT, L THE PROMPT LENGTH, K THE ORACLE SIZE OF EXP#7. THE LOOP’S BUDGET WINS WHEN IT IS BELOW ITS MANDATORY SET (7 BLOCKS ON ONE PROMPT PER MODEL).

rule	v2 collector	masked loop	physical reference
attention	head/token mean; ragged block over members	head-group/query mean; ragged block padded	head/query/token mean
EMA	decay 0.9 from first decode value	H2O-style: last-32 sum; reconstructed: last-8 EMA, 0.9	decay 0.9; last-64 prefill queries
within view	$\bar{a}/(\max \bar{a} + 10^{-9})$	$\bar{a}/(\max \bar{a} + 10^{-9})$	$\bar{a}/\max \bar{a}$
cross view	previous-layer EMA top- $[0.1n]$; layer 0: own EMA	same rank rule; layer 0: own EMA	same rank rule; layer 0: zero
query / age	post-RoPE q ; age from block creation	query 0.5; age since last top- k	unused
label	future top- $[0.1n]$, real lookahead	current unmasked top- K (Exp#7)	unused
budget	—	$\text{round}(0.2n)$, every 8 steps	$\lfloor 0.2 \lceil L/32 \rceil \rfloor$, prompt-fixed
mandatory	—	reconstructed: 4 sink [2] + 4 recent; H2O-style: half heavy, half recent	1 sink + 1 recent
re-admission, new tokens	—	allowed at each decision; new tokens unmasked, budget grows	irreversible; new tokens eligible, fixed budget

left to the artifact. Every fit, including the isotonic control’s calibration split, uses held-in requests only, and no held-out request enters a fit or a threshold; the two-view subset was fixed from version-1 diagnostics, so the version-2 relevance estimates below are descriptive.

A. Exp#1: Ranking and calibration

Version-2 corpus. Table II uses checksum-verified traces of the same 128 LongBench QA prompts (64 Qasper, 64 NarrativeQA) for Llama-3.1-8B and Qwen2.5-7B: 30.4M rows and 256 model-requests. Pooled metrics use 120K training and 150K held-out rows; per-decision metrics use every held-out row. Refitting both models on the same split, seed and test rows at up to all 22.8M held-in rows (190-fold) lifts LightGBM by 0.0005 AUC (0.8961 to 0.8966), widens the paired deficit only from -0.0057 to $-0.0062 [-0.0073, -0.0051]$, and its per-decision recall holds at 0.615, so the gap is stable to training size; six LightGBM capacities (depth 3–8, 150–600 trees, balanced or not) at 120K and 1.92M rows keep its AUC lead between 0.0005 and 0.0076 and its per-decision recall between 0.615 and 0.622, never reaching the raw EMA’s 0.659. A source-prompt split holding out the same 32 prompts for both models agrees with the request-level AUCs within 0.006 for every method: a control, not proof of independence. The two-view fit reaches AUC 0.890 (source-prompt bootstrap $[0.882, 0.898]$) against LightGBM’s 0.896, a paired deficit of $-0.0057 [-0.0068, -0.0046]$, larger than the version-1 -0.0010 but still small. At the per-decision operating point the ordering changes: the raw within-layer EMA recovers 0.659 of

each decision’s future top-decile blocks against 0.613 for the two-view fit and 0.615 for LightGBM, though its pooled top-decile precision (0.542) and AUC (0.873) are lower. Over the 64 held-out requests the paired gaps are +0.046 [0.043, 0.049] and +0.044 [0.041, 0.047], positive on every request (10,000 joint draws of 57 source prompts; +0.047 [0.044, 0.050] and +0.045 [0.043, 0.048] on the source-disjoint split). The reversal’s size depends on the operating point: at the closed loop’s 20% budget, per-decision recall is 0.840, 0.838 and 0.830 and the gaps fall to +0.002 [0.002, 0.003] (52/64 requests) and +0.010 [0.009, 0.011], consistent with, but not explaining, Exp#8’s mean-F1 equivalence.

Where the reversal comes from. Within one decision the within-layer view is a monotone rescaling of the raw EMA (its denominator $\max_{b'} \bar{a}_{\ell, b', t}$ is shared by the candidates), so $\sigma(w_1 x_{w1} + w_2 x_{c1} + b)$ reorders the raw view through exactly one term, the *binary* cross-layer indicator: a fixed offset $w_2/w_1 = 0.025$ ($w_1 = 97.7$, $w_2 = 2.46$) against a corpus mean x_{w1} of 0.015. The lower block of Table II splits pooled AUC exactly on every held-out row. Same-decision pairs are 0.0016% of all (positive, negative) pairs, so pooled AUC is a cross-decision statistic: the indicator raises it by +0.017 [0.015, 0.019] but lowers pair-weighted within-decision AUC by 0.0025 [0.0023, 0.0028] and top-decile recall by 0.046; Fig. 2(a) shows one eighth of the offset already gives 83% of the first and 65% of the second. At the top-decile boundary the term swaps in 22% of the selected blocks, all previous-layer-hot: 20% are positive, against 41% of the blocks they displace. The indicator is informative overall (61% of previous-layer-hot blocks are future top-decile blocks, 4% of the rest) but not at the margin; every decision holds exactly $\lceil 0.1n \rceil$ positives, so it cannot be flagging salient-rich decisions, and why it makes scores comparable across decisions is untested. LightGBM moves the same way (+0.022 pooled AUC, -0.005 within-decision AUC, -0.044 recall).

Relevance and reliability: the magnitude views carry $I(X_i; Z)$ of 0.104 and 0.099 nats, the query proxies 0.007 (cosine) and 0.025 (dot-max), the age proxy under 0.001, on a 400K-row sample of all pooled rows (from held-in rows only they move by at most 0.0011). The class-balanced LightGBM has ECE 0.208 (Fig. 2(b)) while the unweighted (0.005) and isotonic (0.003) variants match the two-view fit (0.006). The isotonic control reserves 48 of the 192 training model-requests, its trees use the other 144, and the 64 test requests are excluded from both fitting stages: request-level calibration separation, not source-level independence.

VI. PHYSICAL-KV REFERENCE AND RUNTIME SCOPE

Scope. Two boundaries hold for every runtime number in this paper. (i) Masking is not physical eviction: equal outputs under physical eviction would need matched token histories, kept sets, position semantics, feature availability and re-admission rules, none of which the simulator establishes, so every closed-loop task-quality result is scoped to the masked loop (and a logical budget, §II, releases no device memory). (ii) Nothing here is a serving result: scoring microseconds,

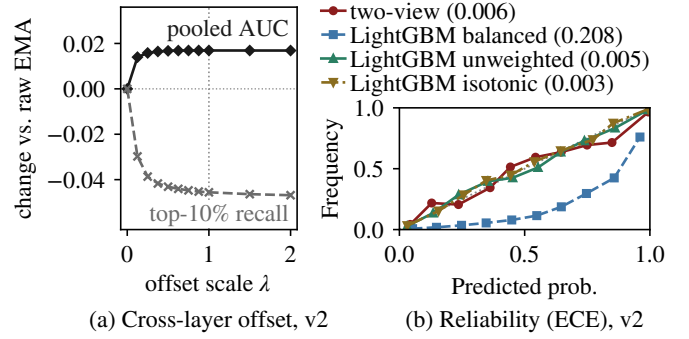


Fig. 2. Exp#1, version-2 corpus. (a) The fitted cross-layer offset scaled from 0 (raw within-layer EMA) to twice its value, on every held-out row: pooled AUC and per-decision top-decile recall as changes from the raw EMA. (b) Reliability of the two-view fit and of class-balanced, unweighted and isotonic LightGBM in 10 equal-width bins; ECE in parentheses.

TABLE II

EXP#1: VERSION-2 RANKING AND CALIBRATION, HORIZON 4, REQUEST SPLIT. UPPER BLOCK: AP IS TIE-AWARE AUPRC; DEC. IS TOP-DECILE PRECISION (EQUAL TO RECALL THERE), MACRO-AVERAGED OVER CACHE DECISIONS; ECE ONLY FOR PROBABILITY OUTPUTS. LOWER BLOCK, EVERY HELD-OUT ROW (7.6M ROWS, 61,184 DECISIONS): POOLED AUC (LIGHTGBM .895 AGAINST .896 ON THE 150K SAMPLE), AUC OVER SAME- AND CROSS-DECISION PAIRS, PER-DECISION RECALL AT 10% AND 20% RETENTION. FURTHER ROWS ARE IN THE ARTIFACT.

method	AUC	AP	P@k pool	P@k dec.	ECE
age proxy	.502	.102	.102	.208	—
query proxy (cosine)	.599	.135	.147	.415	—
prev-layer indicator	.783	.412	.609	.610	—
within-layer EMA	.873	.609	.542	.659	—
LightGBM, balanced	.896	.674	.626	.615	.208
two-view logistic, refit	.890	.667	.624	.613	.006
two-view, archived weights	.890	.667	.622	.611	.019
method	pooled	same	cross	rec. 10%	rec. 20%
within-layer EMA	.873	.930	.873	.659	.840
two-view logistic, refit	.890	.928	.890	.613	.838
LightGBM, balanced	.895	.925	.895	.615	.830

kept-set sizes, local latencies and allocator peaks measure none of production per-token latency, GPU-memory release, contention or throughput, and the reference runner implements no paging, offload or batching.

Exp#6: Physical-KV implementation check. An eager, single-request reference backend with original token positions, irreversible block eligibility and independent compact K/V storage is compared with a dense masked reference under the same eligibility rule (Table I); gates compare native full-cache logits with the no-eviction path and replay a frozen teacher schedule between mask and compact storage. H2O-style here means a shared-head block accumulator, not H2O.

Table III reports both models on one A100-80GB, Transformers 4.51.3, greedy decoding, the same 128 English QA prompts (64 Qasper, 64 NarrativeQA), input cap 4096, at most 128 new tokens and middle truncation on 104/128 prompts. Llama-3.1-8B-Instruct runs in bfloat16; Qwen2.5-7B bfloat16 parity exceeded the preset tolerance (maximum absolute error 0.25 versus 0.15 plus 2% of the logit) while argmax agreed,

TABLE III

EXP#6 (PHYSICAL-KV CHECK) AT 20% BLOCK BUDGET. STORED: MEAN UNIQUE K/V TENSOR STORAGE (MiB) AFTER THE LAST STEP; LIVE: MEAN RETAINED PAYLOAD (MiB).

realization	Llama, bfloat16			Qwen, float32		
	F1	stored	live	F1	stored	live
full physical	.223	496	496	.246	434	434
H2O-style mask	.183	496	94	.233	434	82
H2O-style physical	.181	94	94	.233	82	82
reconstructed mask	.177	496	94	.232	434	82
reconstructed physical	.178	94	94	.232	82	82

so Qwen is reported in float32. At 20% budget compact storage equals the live payload whereas the mask keeps full-size tensors, a $5.3\times$ ratio for both. Greedy sequences match between the two paths on 98/128 and 100/128 Llama requests (mean absolute F1 differences 0.014 and 0.007) and 128/128 Qwen requests. Allocator decode peaks fall only from 15.51 to 15.10 GiB (Llama) and 29.05 to 28.63 GiB (Qwen) because weights dominate, with prefill-plus-first-compaction peaks at 15.77 and 29.23 GiB. These free-running agreements are the weakest of three parity levels: the teacher-forced replay of §IX passes at 1024 tokens and four steps but exceeds its tolerance over 40 evictions, so the table supports the storage observation, not serving equivalence.

Why this table’s F1 is half of Table IV’s. This table uses only the two hardest LongBench QA sets where Exp#8 averages seven, Exp#8 scores against every reference (alone lifting the same 896 outputs from 0.355 to 0.419), and decoding and truncation differ: compare within a table only.

VII. CLOSED-LOOP TASK QUALITY

Exp#8: Matched-retention answer quality. The policy evaluated here is the runtime reconstruction of §IV-B, not the offline predictor of Table II. The main comparison uses chat-formatted LongBench QA [10]: its seven English question-answering datasets (NarrativeQA, Qasper, MultiFieldQA-en, HotpotQA, 2WikiMultihopQA, MuSiQue, TriviaQA; abbreviated in Fig. 3), two architectures (Llama-3.1-8B and Qwen2.5-7B) and 448 source prompts, 64 per dataset, each evaluated on both architectures (896 model-prompt evaluations, 64 per architecture-dataset cell), at 20% logical retention, each policy driving its own masked decoding trajectory.

Inferential unit and margin. Because the two architectures answer the same questions, the 14 cells are not independent. The primary analysis averages the two models within each dataset and runs paired TOST over the seven datasets (df 6); the 14 cells treated as independent are a sensitivity analysis. The margin is two absolute F1 points, 5.4% of the rerun baseline 0.372: a practical tolerance carried over from the archived analysis, not pre-registered and not a deployment threshold. Fig. 3 exposes task heterogeneity: mean equivalence is not per-task.

Pinned rerun. With the official LongBench scorer and all references, the reconstructed scorer’s mean F1 difference from

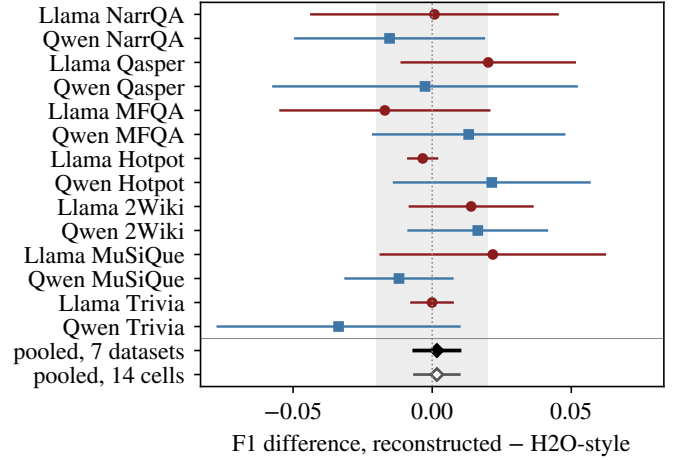


Fig. 3. Exp#8, pinned bfloat16 rerun. Reconstructed-minus-H2O-style F1 per architecture $\times$ dataset cell (64 paired requests, 95% t intervals; the two models of a dataset adjacent) and pooled estimates (90% t intervals): seven datasets (primary, filled) and 14 cells (sensitivity, hollow); the band is the ± 0.02 margin.

TABLE IV

EXP#8: MASKED-LOOP LONGBENCH F1 AT 20% RETENTION, 896 MODEL-PROMPT EVALUATIONS. Δ IS THE DIFFERENCE FROM THE H2O-STYLE ACCUMULATOR WITH A 90% t INTERVAL OVER THE SEVEN DATASETS (DF 6); p_{TOST} IS THE TOST p AT ± 0.02 OVER THE SAME DATASETS. PINNED BFLOAT16 RERUN, OFFICIAL SCORER, ALL REFERENCES.

policy realization	F1	Δ [90% CI]	p_{TOST}
full cache	.419	+0.047	—
H2O-style	.372	—	—
reconstructed	.374	+0.002 [-0.006, +0.010]	.0023
Ada-KV-style	.363	-.008 [-0.016, -.000]	.015
PyramidKV-style	.349	-.023 [-0.046, +0.001]	.582

the accumulator over 896 evaluations is +0.0017: the seven-dataset TOST gives $p = 0.0023$ at ± 0.02 and the 14-cell sensitivity analysis $p = 0.0007$. The seven-dataset t interval is $[-0.0064, 0.0098]$; the 14-cell t -based and cell-bootstrap 90% intervals are $[-0.0063, 0.0097]$ and $[-0.0056, 0.0087]$. Supplementary ± 0.01 tests ($p = 0.047$ seven datasets, 0.045 14 cells) use a margin that was not set in advance, so we do not count them. Per architecture the difference is +0.0052 (Llama, $p = 0.016$) and -0.0018 (Qwen, $p = 0.026$). The Ada-KV-style comparator remains equivalent (-0.0084 , $p = 0.015$); the PyramidKV-style one does not (-0.023 , $p = 0.58$). Both allocation comparators only move a fixed total budget across layers (Ada-KV-style: by attention concentration; PyramidKV-style: a linear pyramid) and select within a layer as H2O-style does, so no finding covers original H2O, Ada-KV or PyramidKV, or equivalence between the reconstructed and Ada-KV-style policies; and this is equivalence between compressed policies, not absence of compression loss: both stay 0.045–0.047 below full cache. The policies also differ in EMA window and mandatory blocks (Table I): this compares two complete runtime realizations, not a scorer-only intervention,

which is untested. Logged retained blocks per decode step average 25.02 for the H2O-style, reconstructed and Ada-KV-style policies and 25.01 for the PyramidKV-style one (median 26, range 7–26): retention is matched in counts, not only in the nominal 0.2.

What the rerun repaired. Archived Qwen float16 decoding emitted all-NaN logit rows, producing token 0 (“!”): a 56-prompt replay found this in 54 requests, against none in bfloat16/float32. In the rerun no Qwen full-cache output contains a run of “!” (0/448 versus 181/448) and 34/448 reach the 48-token cap (versus 210/448). The archived F1 difference was +0.00511 under a single-reference scorer (+0.0030 after all-reference rescaling); the artifact keeps the original scores and the repair record.

Exp#9: Archived budget and context checks. Sweeping retention over {0.10, 0.20, 0.30, 0.50} yields no resolved per-budget quality gain; at 16K the larger Llama sample (448 prompts, 20% retention) has mean F1 difference −0.0006, request-level 95% interval [−0.012, 0.012], and a 160-prompt pilot that had suggested +0.010 is not sustained, so we claim no long-context gain. A synthetic needle probe is a negative control, not evidence about query-aware selection. Neither check proves curve equivalence or zero memory saving.

VIII. HISTORICAL DIAGNOSTICS

Every measurement here comes from the unpinned version-1 pipeline, whose defects—float16 Qwen2.5 NaN logits, a query extractor that never formed the attention-space product, an AUPRC that credited tied scores for row order—are in §IX. Its raw rows are gone, so none of these numbers is version-2 evidence and none can be recomputed; we keep them because they motivated the audit, and docs/HISTORICAL_DIAGNOSTICS.md lists every one with its source; identifiers follow the artifact’s order.

Exp#1–5: offline, thresholds and cost. The corpus holds 43,931,784 rows from four instruction-tuned families; its two-view AUC is 0.948 [0.942, 0.954] against LightGBM’s 0.949 (paired deficit −0.0010 [−0.0016, −0.0003]), the ordering the corrected corpus shows at a wider margin. Workload extensions keep that ordering and cross-model transfer loses little (their AUPRC used the row-order tie rule and the rows are gone, so we do not restate it); within a generation a frozen scorer keeps AUC 0.948 while the online IRLS update reaches 0.919, evidence against that update, not against online learning. GuardKV—our name for a risk-target thresholding rule, not a published system—tracks its target miss within 0.007 in distribution, with retention near 34% and 14% at targets 0.05 and 0.20, but against the served-oracle target the reconstructed scorer needs about full retention for 90% coverage, so version 2 does not build on it. Scoring one 4096-vector batch with the four-weight kernel takes 28.7 μ s at P99.9, excluding extraction, selection and compaction.

Exp#7: Offline and in-loop targets differ. The archived offline two-view AUC is 0.948, while on the served-oracle calibration set the reconstructed scorer reaches 0.643 ([0.632, 0.655], per-layer 0.54–0.76). We report the drop but

do not rely on it: the bridge of §IV-B reproduces neither its size nor its direction on version-2 traces, and target, trajectory and feature realization change together here, so its cause is unidentified. On four LongBench tasks at 20% retention (48 requests each) the H2O-style accumulator has request-mean served-oracle miss 0.579 against 0.623 for the reconstructed scorer (paired difference [0.033, 0.055]) while the per-task F1 intervals all overlap; the confirmatory comparison is Exp#8. Part of any served-oracle miss is arithmetic, since a budget below the oracle set cannot be met: from logged per-step counts the 4K-cap mean floor is 0.219, leaving residuals 0.360 and 0.404 and a paired residual interval still [0.033, 0.055], so the policy difference survives the correction. The feedback-shift check (a scorer trained on the served-oracle target gains 0.06 [0.03, 0.10] off-policy recall but loses 0.13 [0.10, 0.16] on-policy miss) and the other caps and budget fractions are in the artifact.

IX. LIMITATIONS AND ARTIFACT AUDIT

External validity. Historical traces cover four instruction-tuned 7–8B families; corrected v2 traces and the pinned rerun cover two, at up to about 4K context. The archived 16K Llama and short-context Qwen2.5-14B runs are separate checks, not v2 replications (14B F1 0.301/0.307/0.301 for the accumulator, reconstructed scorer and adaptive allocation). Nothing is extrapolated to MoE, base checkpoints, 32K contexts or sampling; decoding is greedy.

Query-coordinate audit. The legacy extractor hooks `q_proj` before rotary transformation but reads post-rotation keys, so it never computed the attention-space query–key product (its revision is unpinned); the old dot-max probe is no faithful Quest control and we claim no intrinsic query redundancy. Version 2 aligns queries and keys after RoPE, with a one-head test that its scaled dot products reproduce the attention probabilities; even so a dot-max probe is not Quest’s page-bound algorithm. Adding it to the two magnitude views changes held-out *request-mean* AUC by −0.0070 on the 128 Llama prompts (32 test requests, all negative; [−0.0082, −0.0058]; two-view mean 0.899) and by +0.0020 on the matched Qwen prompts (31/32 positive; [0.0016, 0.0024]; mean 0.884): small, model-dependent, opposite in sign.

Generation, labels and the archived loop. The legacy collector refed the final prompt token and clamped unavailable horizons to the last recorded step; version 2 consumes the prefill-predicted token and generates genuine lookahead, so archived long-horizon results are not evidence for long-horizon generalization (no archived number was silently changed). The archived cells came from an unpinned float16 simulator whose Qwen2.5 path emits all-NaN logit rows (§VII), scored against one reference with articles removed before punctuation; the rerun pins the revision by hash, uses bfloat16 for both architectures and applies one memory patch (each layer’s already-ingested attention weights replaced by a placeholder in the tuple returned to the decoder layer), verified on a tiny model to change no statistic, mask or token. No rerun latency is reported; the GPUs were shared.

Tied scores and shared prompts. The archived AUPRC integrated one row at a time after a stable sort, crediting tied scores for row order; the two discrete probes cannot be recomputed, so Table II uses the threshold-based definition. The archived split held out model-requests, not source texts; version 2 adds a source-prompt split.

Inference, aggregation and targets. Information estimates depend on binning and conditional sparsity; overlapping marginal intervals are not an equivalence test; the primary TOST treats the seven dataset means as independent units, so it holds for this fixed collection of two architectures and seven tasks, exploratory samples do not replicate it, and non-rejection is not proof of no benefit. The collector averages attention with equal weight over all query heads and over a block’s actual members, whereas the loop averages head groups and pads ragged tail blocks (Table I), a difference we did not measure. The label is the attention four steps ahead while the loop re-decides every eight steps; a window target is untested. Both learned scorers minimize a pooled loss (a per-decision ranking objective is untested); a budget shared across layers or requests makes cross-decision ordering the operating point; and offline candidates include sink and recent blocks the loop always keeps.

Parity levels of the reference backend. The committed gates (1024 tokens, four decode steps) pass their numeric tolerance. Extending them to four prompts from both datasets, 1024–4096 input tokens and 40 decode steps with repeated eviction separates three levels: the session reproduces native full-cache logits exactly (maximum error 0 in all 12 runs); but the compact and masked paths, fed the same teacher tokens and schedules, drift apart over successive evictions to a maximum logit difference of 0.66–1.44 in bfloat16 ($3.5\text{--}5.9\times$ the preset tolerance) and 3.6×10^{-5} – 1.9×10^{-4} in float32 ($1.3\text{--}6.1\times$), so the extended test fails that tolerance. The four-order fall with precision fits floating-point ordering over 40 evictions but does not exclude a semantic difference; settling it needs a layer-by-layer comparison of retained indices, positions and K/V values against a higher-precision reference. Top-1 agreement is 972/984 gated steps in bfloat16 and 984/984 in float32. Tolerances were not raised. The old physical-memory probe, which swallowed decode exceptions and put full prefill in its peak statistic, is retired, its source and results kept.

Availability. The artifact contains LaTeX, result JSON, analysis code, predictor weights, tests, the version-2 collector, the reference validator and the rerun’s exact simulator source, bundled with its MIT notice and verified against all 70 cells’ provenance hashes, plus a CPU example; model weights remain separately licensed. Code, records and figure sources are at <https://github.com/lzbaclz/KVSaliencyBench>; the default benchmark entry scores a prediction table without traces or weights, and raw traces are available on request.

X. RELATED WORK

Streaming learning studies evolving examples and distribution shift [11]; multi-view learning and relevance–redundancy analysis motivate comparing information sources [12], [13],

which we apply to recorded attention dynamics without treating a small information estimate as a sufficiency proof. Our pooled-versus-within-decision split parallels user-weighted (grouped) AUC in click-through prediction [14], which averages AUC within users because pooled AUC also rewards cross-user order. Calibration and conformal prediction [15]–[17] provide tools whose guarantees do not transfer to dependent runtime observations unchecked.

KV-management methods use heavy hitters, attention sinks with recent windows, recent observations, cross-layer signals, query bounds, adaptive allocation and learned retention [1]–[6], [18], [19]; PagedAttention [20] addresses physical serving. Our probes and allocation comparators are not interchangeable with those systems; this is an audit of the links between intermediate prediction and task quality, not a claim that these methods are unnecessary.

Closest to our question are prediction-first methods and evaluation suites for KV compression. AttentionPredictor [7] learns a lightweight model of next-token attention and uses it to identify critical tokens; we do not compare against it, and ask a different question: which of pooled ranking, the per-decision budget, runtime features, the candidate set and task quality a proxy’s ranking survives. Gao et al. [21] audit compression from the serving side (unmeasured throughput costs, longer outputs, per-sample quality variation hidden by averages), and Agrawal and Mayer [22] benchmark task quality, throughput, time-to-first-token and compression ratio jointly. SCBench [23] evaluates the whole KV-cache life-cycle, including shared-context reuse, and reports attention-distribution shift in long generation; KVPress [24] standardizes more than twenty compression methods; KVzap [25] is a recent learned pruning method evaluated on such a suite, and Locret [6] trains retaining heads on a causal-importance target and validates them by end-task quality. Task-level benchmarks such as LongBench and RULER [10], [26] score answers; KVSaliencyBench records intermediate features and attention-derived labels, a complementary level: those suites fix a common harness and ask whether a method is worth deploying, whereas we fix the scorer and the label and vary the feature realization and the candidate set, to ask which link a proxy ranking survives (§IV-B, Exp#7–8). The negative findings concern how proxies transfer to task quality in the tested implementations, not the value of learned or value-dependent selection [27].

XI. CONCLUSION

KV-saliency prediction and KV retention are related but distinct evaluation problems. On version-2 traces, the ranking advantage of learned scorers in pooled metrics reverses under per-decision top-decile budgets, less sharply at 20% retention. Pooled AUC is almost entirely cross-decision pairs; the logistic scorer’s binary cross-layer term raises it and lowers within-decision AUC and recall, promoting previous-layer-hot blocks that are positive less often than the blocks they displace. A separate pinned bfloat16 masked-loop rerun

finds the reconstructed scorer mean-F1 equivalent to the H2O-style accumulator within a practical ± 0.02 margin over seven datasets, as two complete runtime policies, while both lose quality relative to full cache. The bridge puts reference-backend feature reconstruction within 0.003 AUC; its on-policy arm ranks only surviving blocks, so the archived served-oracle gap stays a historical anomaly for want of a label-only arm. The physical reference shows what storage a compact cache needs, not that it is equivalent: its extended replay exceeds the numerical tolerance. These findings concern the tested data and implementations, not the impossibility of better query-aware selection. To evaluate a retention scorer, fix the candidate set and budget, measure inside each decision, then check answer quality on the policy’s own trajectory; the artifact’s default protocol does the first two.

ACKNOWLEDGMENT

Anthropic Claude (via Claude Code) assisted with manuscript drafting and the version-2 collector, physical-KV validator, audit scripts and decision-level benchmark protocol. OpenAI ChatGPT/Codex assisted with revisions to the abstract, introduction, evaluation diagram, statistical protocol, results text, limitations and conclusion, and with bootstrap, oracle-budget, reproducibility and PDF-check code. These tools generated or restructured text and code; the authors are responsible for verifying the retained content, experiments and conclusions.

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